

Koa Shen

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Hiring Team

SLAC National Accelerator Laboratory

Menlo Park, CA

Re: Returning Mechanical Engineering Internship, Summer 2027

Dear SLAC Hiring Team,

I am writing to express my interest in returning to SLAC National Accelerator Laboratory for a mechanical engineering internship in Summer 2027. My work with LCLS has shown me how much I value engineering in a research environment where careful mechanical design directly supports reliable experiments.

This summer, I built a Python digital-twin framework for simulating complex X-ray beamline motion assemblies, with live collision detection and a reusable catalog of motion stages. The framework is now being applied to multiple LCLS assemblies. I have also used it to guide a redesign of the polycapillary optics assembly, translating simulation findings and instrument-scientist feedback into practical changes to mounting, alignment, and detector positioning. Continuing this work would let me build on a system and team I already know while helping move the designs toward dependable use.

My contribution has spanned more than simulation: I have produced mechanical designs and PDM-controlled documentation in Siemens Solid Edge, applied GD&T, and performed statics and dynamics checks for sensitive equipment. At UC Santa Barbara, I study mechanical engineering and lead fabrication work with Gaucho Racing, where I design and machine hardware for a student-built electric race car. These experiences have taught me to carry engineering decisions from analysis through manufacturing and hands-on validation.

I would be glad to return to SLAC and continue contributing to LCLS motion-system design and simulation while taking on new mechanical engineering challenges. Thank you for your consideration.

Sincerely,

Koa Shen